Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/30/2011
PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	Coarsening in Solid-Liquid Mixtures (CSLM)-2: NNX07AW01G		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-0893	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2007	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 9/30/2014 per NSSC information (Ed., 10/25/2012) NOTE: End date is now 9/30/2012 per NSSC information (Ed., 6/7/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AW01G		
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Performance Goal Text:			

Task Description:	Previous NASA sponsored work clearly showed that solid-liquid mixtures consisting of Sn-rich particles in a Pb-Sn eutectic liquid are ideal, and perhaps unique, systems in which to explore the dynamics of the Ostwald ripening process. The high coarsening rate in these systems permit accurate kinetic data to be obtained, and the thermophysical parameters necessary to make a comparison between theory and experiment are known. However, in a terrestrial environment experiments can be performed only at the relatively high volume fractions of solid where the presence of a solid skeletal structure prevents large-scale particle sedimentation. In these high volume fraction experiments, a comparison between theory and experiment shows that the solid-liquid mixtures are coarsening faster than predicted by an approximate theory for purely diffusional controlled Ostwald ripening. The objective of this project is to plan and perform a microgravity experiment on Ostwald ripening in solid-liquid mixtures. This experiment will serve two primary purposes: it will allow experiments to be performed which can be directly compared to heretofore untested theories for coarsening in systems with low volume fractions of solid, and it will eliminate conclusively convection of the liquid matrix and small-scale particle motion within the skeletal structure as possible sources for the disagreement observed between theory and experiment in the high volume fraction experiments. To perform coarsening experiments whose results can be compared directly with theory with no adjustable parameters. The spaceflight experiment will produce data which, for the first time, can be compared directly to theory with no adjustable parameters. This data will address the long standing controversy over the dependence of the coarsening rate of a two-phase system on the volume fraction of coarsening phase.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	CSLM-2 will make possible predictions of coarsening kinetics in alloys with high volume fractions of precipitates, such as alloys under consideration for high temperature exploration applications (i.e., nuclear propulsion, waste heat coolant loops). Results will be applied to materials design codes that will reduce cost, shorten development time, and improve performance over conventional materials design methods.
Task Progress:	We have made significant progress in many areas this past year. We have completed analysis of the 3D reconstruction of the 48-hour sample, and compared the results to theory. We have made 300 sections through a 30% sample coarsened at an earlier time to determine if steady state coarsening was present. We have made two-dimensional sections through samples with 10%, 15%, and 20% volume fractions of Sn. These are the samples that arrived this past spring from the ISS. A brief summary of the results is below. The three-dimensional reconstruction of the 30% sample has yielded the first measurement of the 3D particle size distribution and spatial distribution of system undergoing coarsening with such a high volume fraction. The measured particle size distribution is close to that predicted by theory, closer than any other previously measured particle size distribution. Nevertheless, there are noticeable differences between these two distributions. In order to determine possible causes of this we have computed the spatial distribution function. We find that there is a significant disagreement between theory and experiment. We also find that many of the particles do not have the spherical morphology required by theory. Most have distortions in their shapes that yield ellipsoidal like shapes. These distortions are due to the packing of the particles at this high volume fraction of particles. Consistent with this, an analysis of the 3D reconstruction shows that all the particles are connected in three dimensions. Using the known materials parameters, and the conversion factor from the average particle radius measured on a cross section to that in three dimensions, we find that the mixture is coarsening faster than predicted by theory. We made a second three-dimensional reconstruction of a sample at an earlier time in part to determine if this conversion factor is a constant. We have analyzed the 10%, 15% and 20% samples using 2D sections through the samples. The 10% samples were not successful. The samples with 15% volume fraction yielded very nice microstructures, with spherical non-touching particles. Thus, this volume fraction can serve as a critical test of theory, as is the case with 10% volume fraction. We are still investigating the causes of the unusual microstructures in the 10% volume fraction sample. The 20% sample also yielded good microstructures, with non-contacting spherical particles, and thus these samples can also serve as a test of theory. We find that the cube of the average particle radius as measured on a plane section increases linearly in time and that the rate constant increases with volume fraction, as theory predicts. The particle size distribution functions as measured on a plane section are nearly time invariant when scaled by the average particle radius, and thus it appears as if the system has attained steady state coarsening. It is also clear that the distributions for the two different volume fractions of solid are slightly different, with the particle size distribution at 15% being narrower than that at 20%. To compare the results to theory, it is necessary to measure the particle size distributions in three dimensions.
Bibliography Type:	Description: (Last Updated: 01/17/2024)
Papers from Meeting Proceedings	Gulsoy EB, Wittman K, Thompson J, Voorhees PW. "Coarsening in Solid-Liquid Mixtures: Effect of Microgravity Accelerations on Particle Sedimentation." 49th AIAA Aerospace Sciences Meeting, Orlando, FL, January 4-7, 2011. 49th AIAA Aerospace Sciences Meeting, Orlando, FL, January 4-7, 2011. Paper AIAA 2011-1346. , Jan-2011